Linear Regression

CS 372 Machine Learning I

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Outline

- Big Idea.
- Simple Linear Regression.
- Regression Model's Parameter Estimation.
- Multiple Linear Regression.
- Normal Equations.
- Regression Model Discussion and Evaluation.

office ID	Size	Floor	Energy Rating	Far from downtown	……..	Price $
1	500	4	C	1		320
2	550	7	A	3		380
3	620	9	A	5		400
4	630	5	B	6		390
5	665	8	C	7		385
6	700	4	B	8		410
7	770	10	B	2		480
8	880	12	A	2		600
9	920	14	C	3		570
10	1000	9	B	1		620

office ID	Size	Price $
1	500	320
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9	920	570
10	1000	620
11	501	322
12	556	384
13	622	406
14	635	380
15	660	386
16	701	411

Training dataset

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1	500	320
2	550	380
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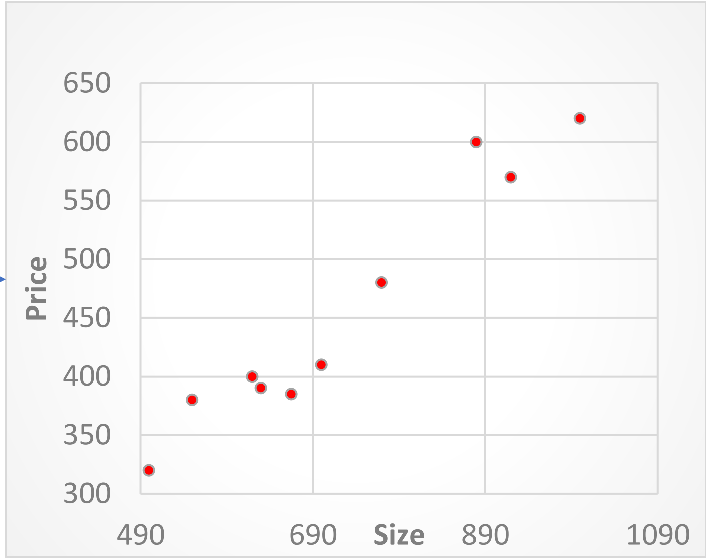

Test dataset

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11	501	322
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13	622	406
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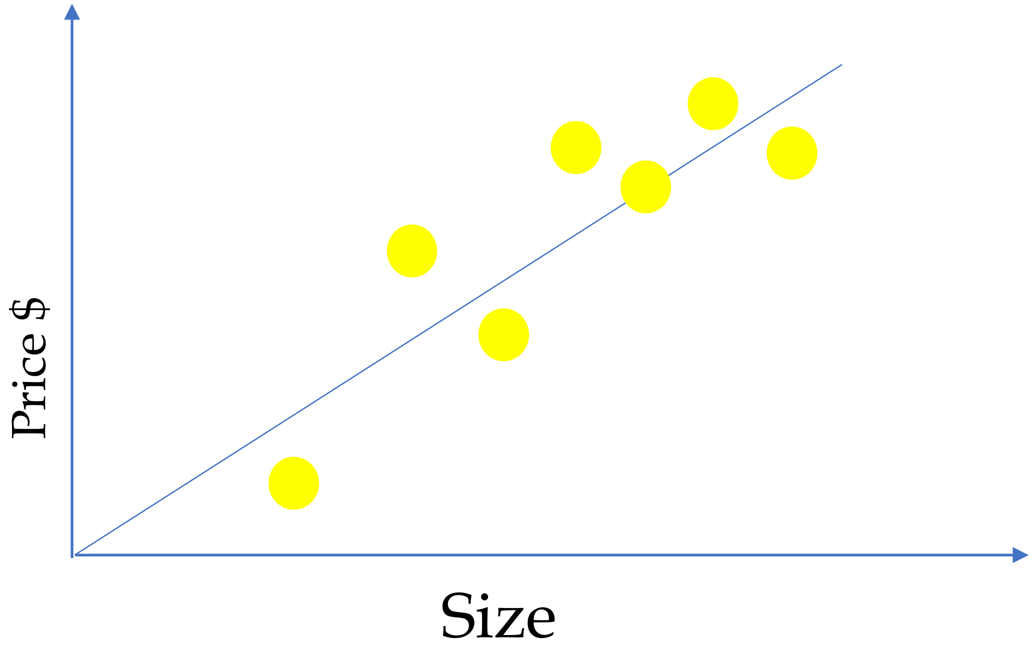
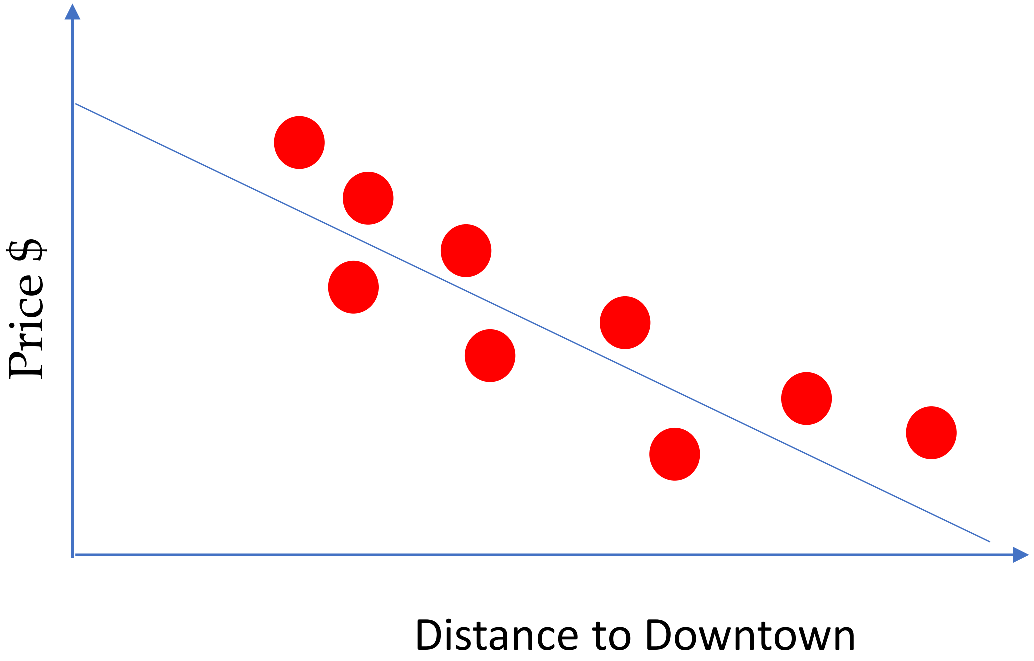
16	730	461

Introduction

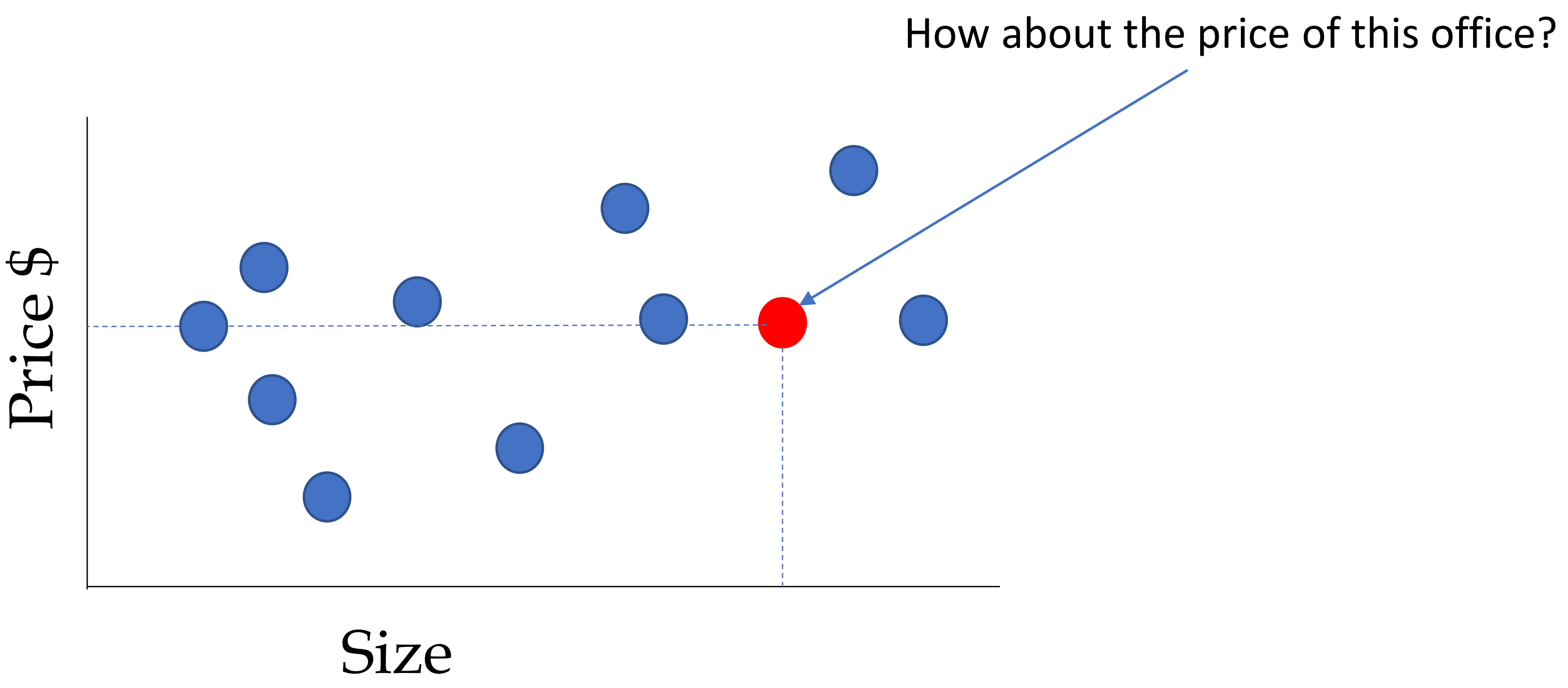
1. Modeling the relationship between one or multiple features and a continuous target variable.
    - Examples: office rental size and price, office rental price and distance to downtown.





Introduction

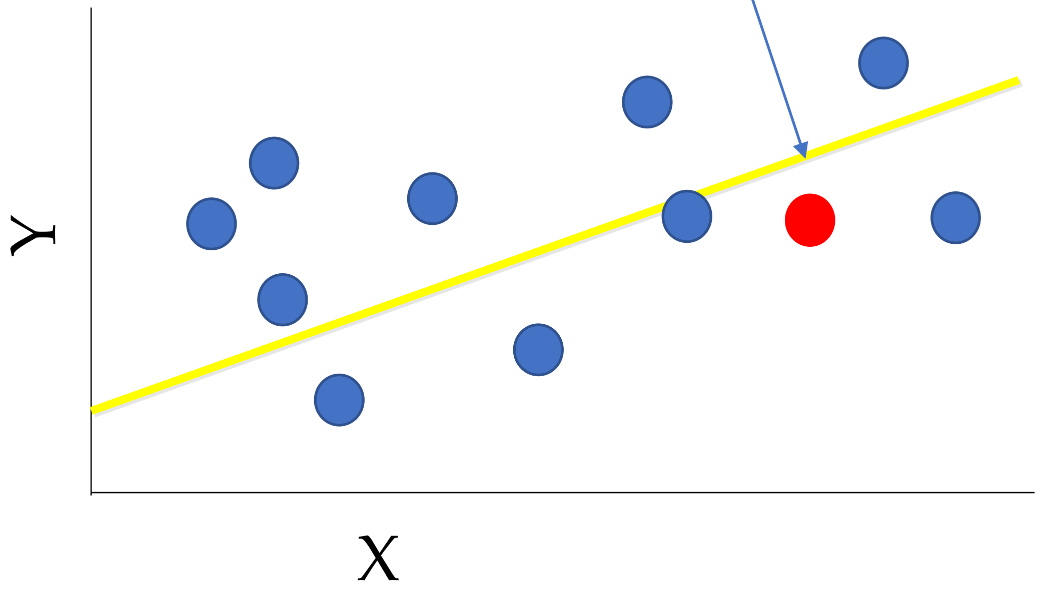
2. Forecast new observation
    - Examples: predict office rental price.



Some Math

- Equation of a line

$$y = a\,\mathrm{x} + \mathrm{b}$$



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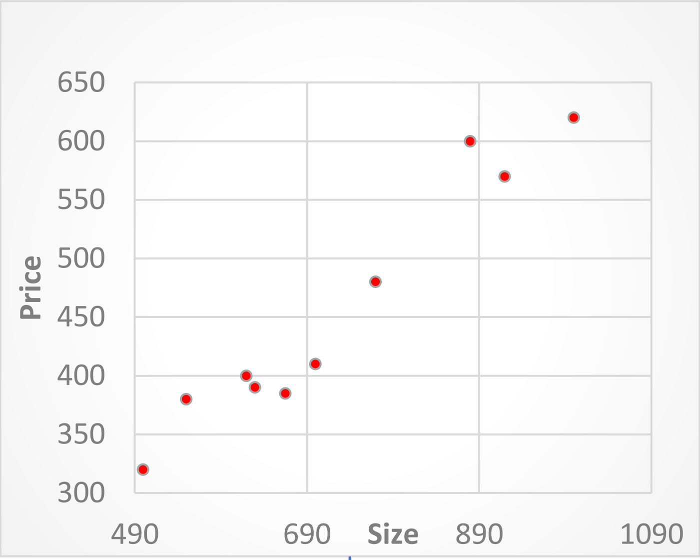

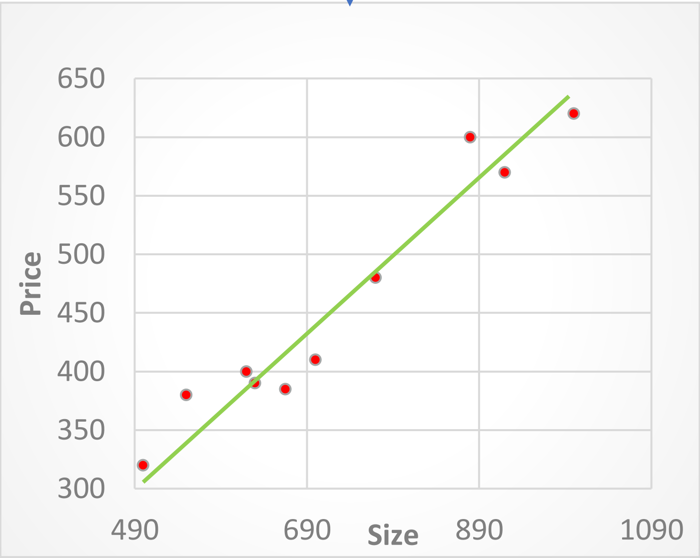

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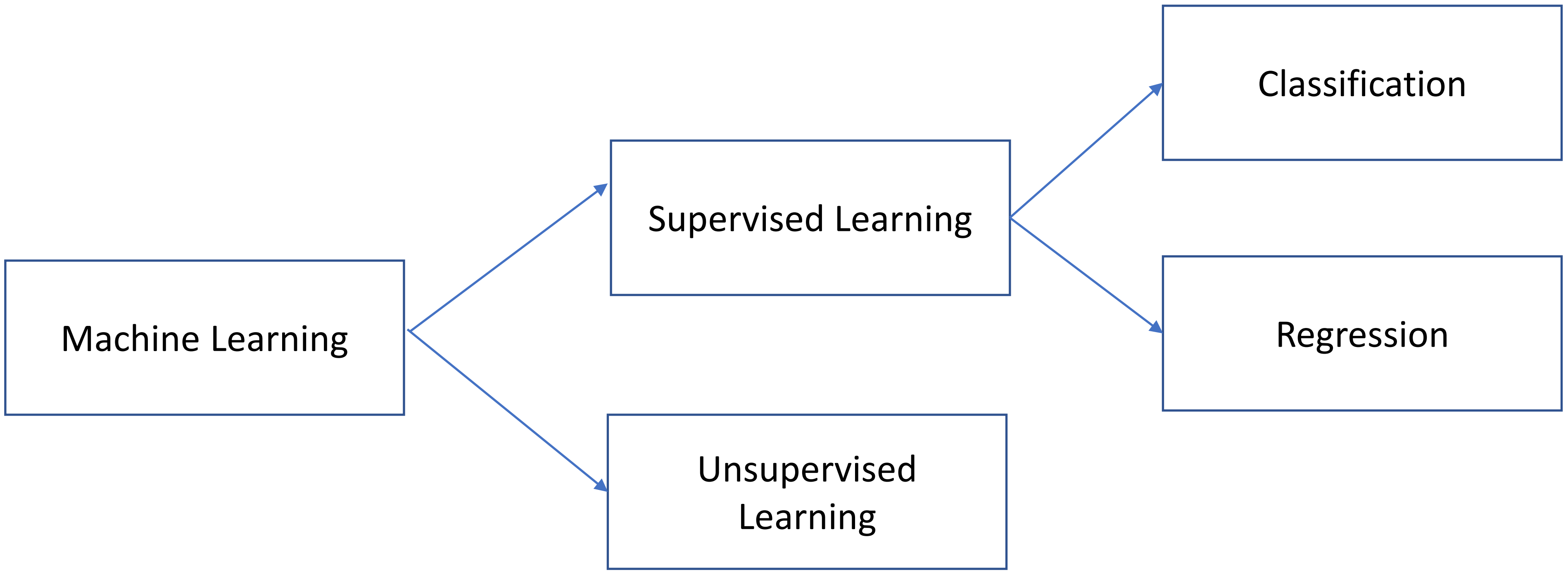
Machine Learning Types



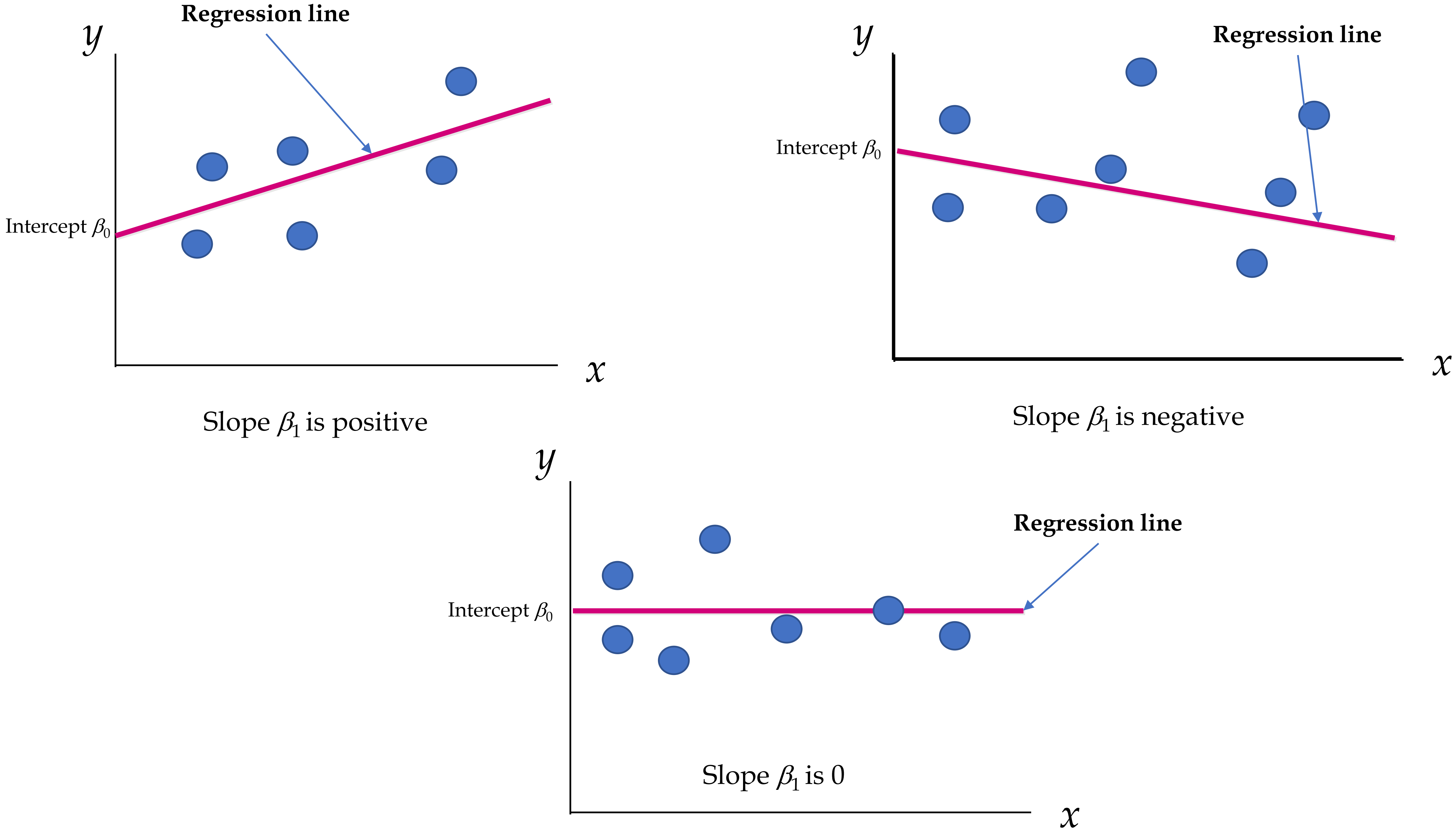
Simple Linear Regression

Given a set of training data (x_1, y_1) … $(\mathrm{x}_\mathrm{N}, \mathrm{y}_\mathrm{N})$, and we want to predict Y (real-value). The linear regression model has the form

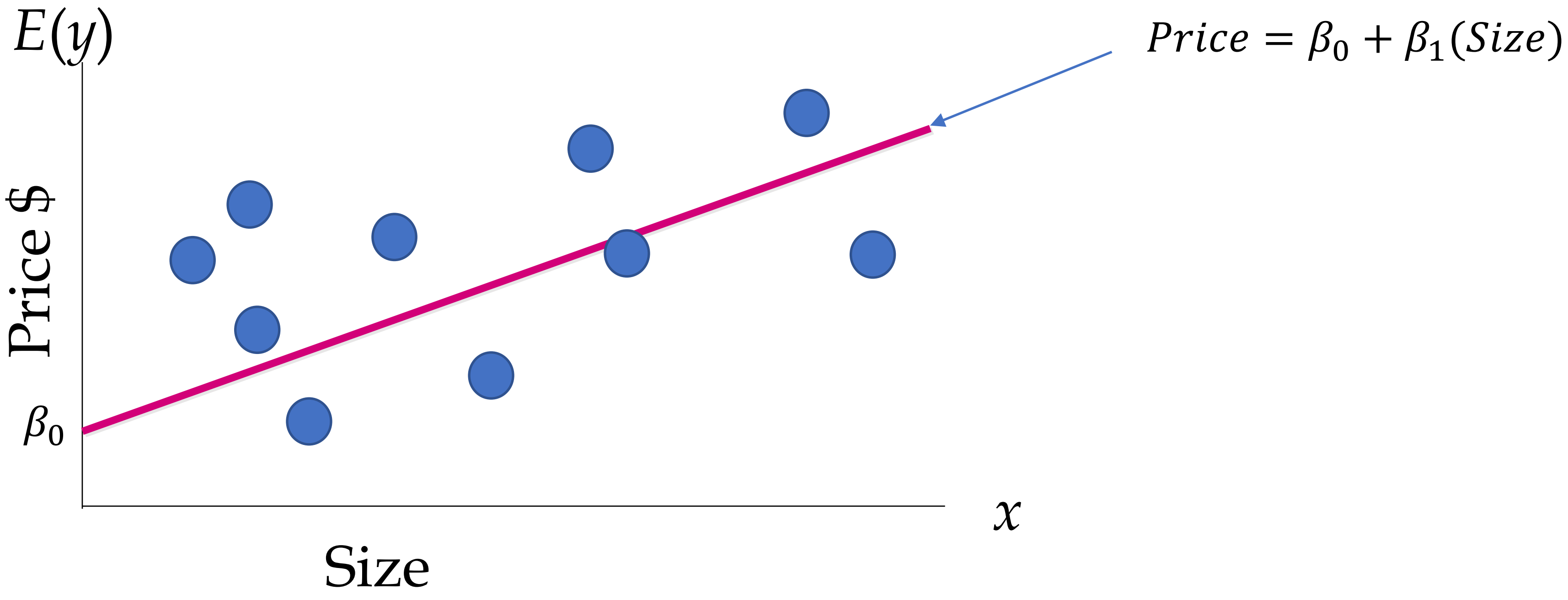
$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- Y is the dependent variable / response variable.
- X is the independent variable/ predictor / explanatory variable
- β_0 is the constant coefficient or Y-intercept
- β_1 is the slope
- ε is the model error.

$$Y = \beta_0 + \beta_1 X + \varepsilon$$



Task: Predicting Office Rental Prices



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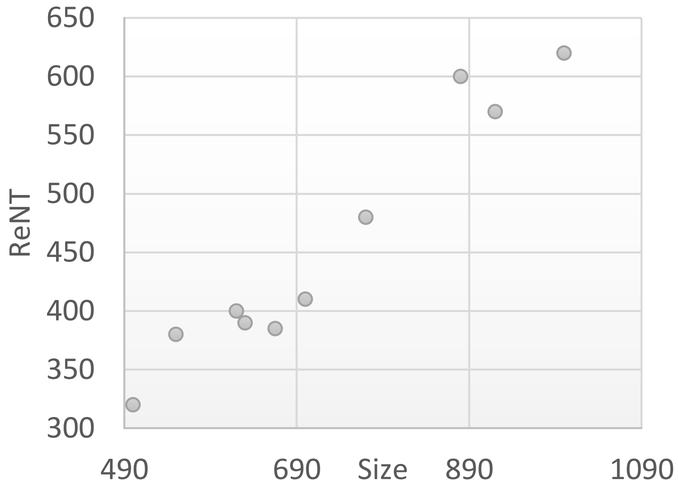
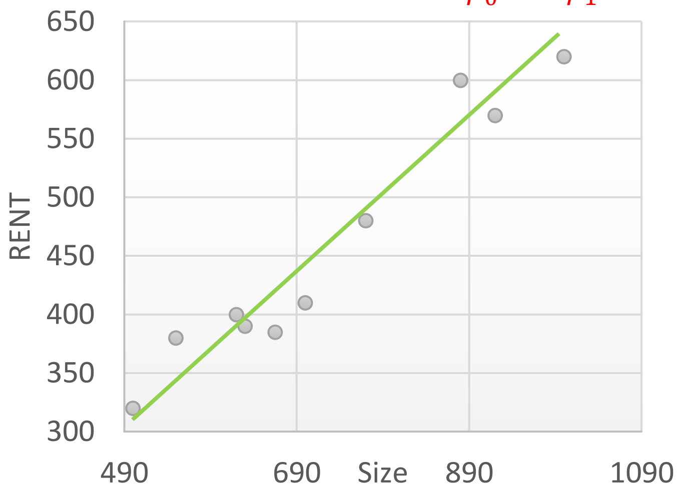
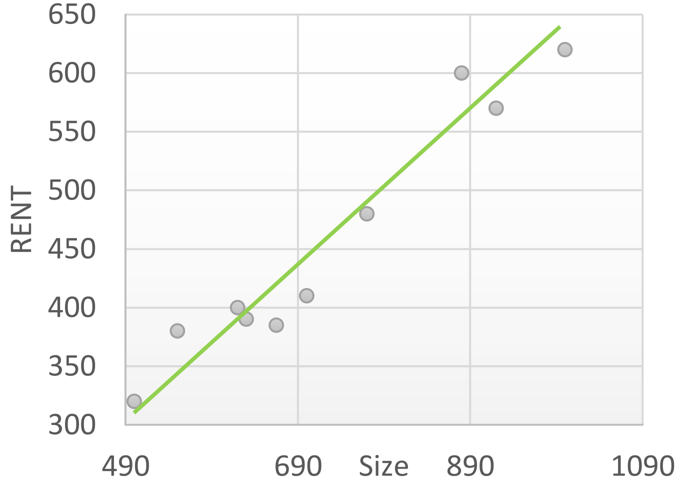


$\hat{Y} = \widehat{\beta_0} + \widehat{\beta_1} X$



Price $= 6.47 + 0.62 (SIZE)



Which Model?

office ID	Size	Price $
1	500	320
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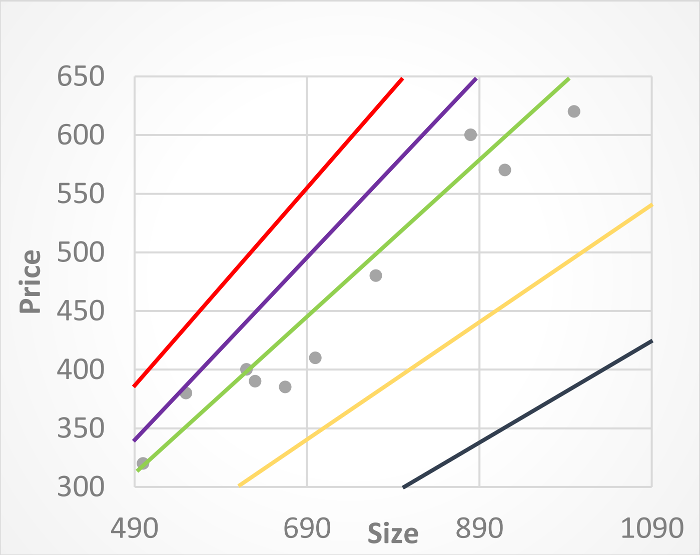
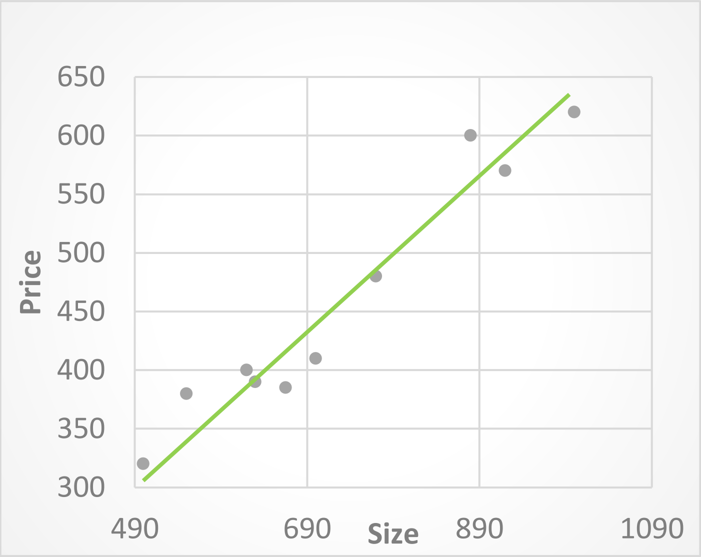


Question: What should be considered a good line?

Solution: The smaller the sum of squared differences the better the fit of the line to the data.

Residual Sum of Square (RSS)



Regression Model

$$Y = \beta_0 + \beta_1 X$$

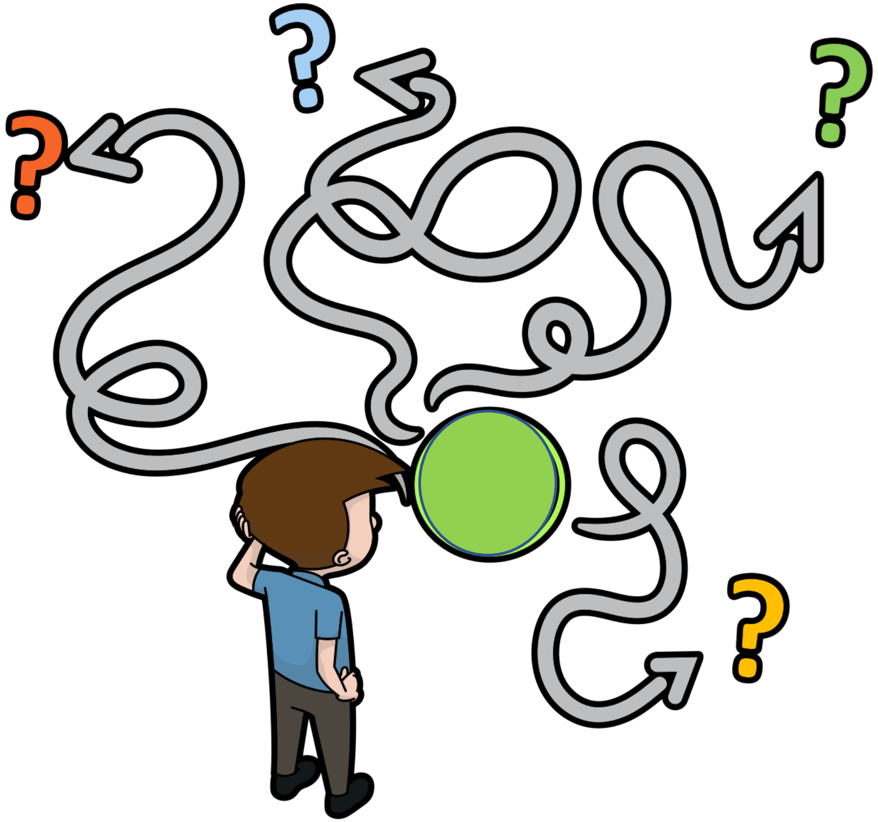
Estimated Model

$$\hat{Y} = \widehat{\beta_0} \quad + \widehat{\beta_1} \quad X$$

- Slope: $\widehat{\beta_1}$
- y-intercept: $\widehat{\beta_0}$

Estimate the slope and intercepts

- Brute-force search
- Analytical Solution
- Estimation



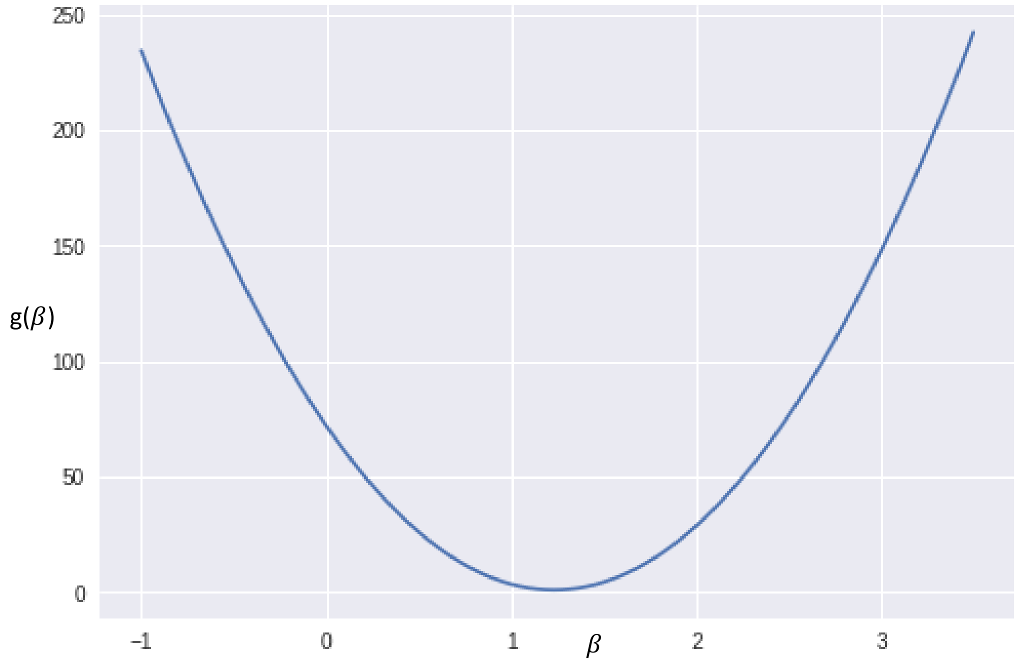
Least Squares Method

Least Squares Criterion to estimate regression parameters

$$RSS(\beta_0, \beta_1) = \min_{\beta_0, \beta_1} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

where:

- y_i = observed value of the dependent variable for the ith observation.
- $\hat{y}_i$ = estimated value of the dependent variable for the ith observation



How to estimate Regression line coefficients?

- Regression Model

$$Y = \beta_0 + \beta_1 X$$

- Estimated Model

$$\hat{Y} = \widehat{\beta_0} \quad + \widehat{\beta_1} \quad X$$

Slope

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

***y*-intercept**

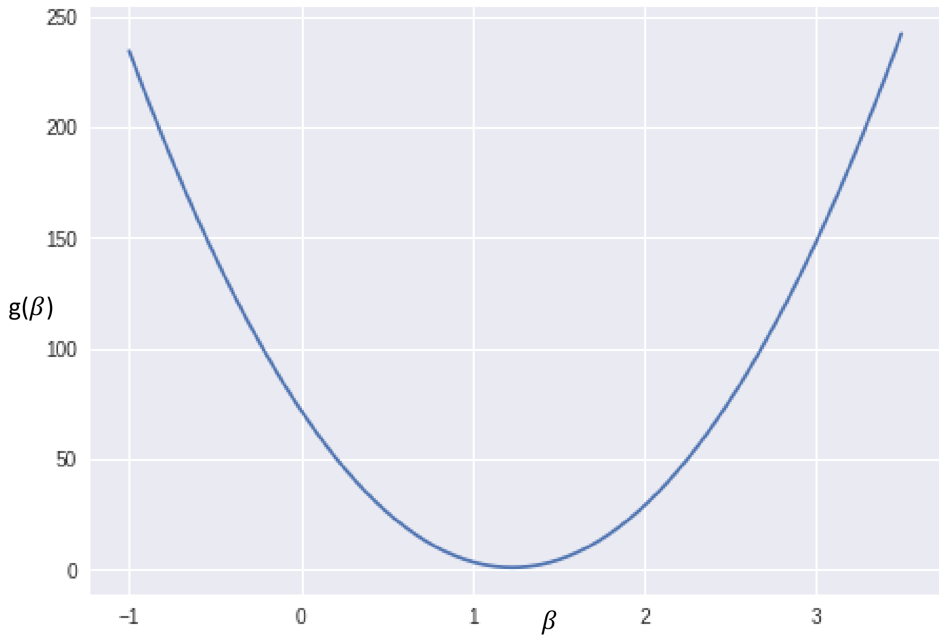
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Gradient Descent Algorithm

- While not converged

$$\beta_j^{t+1} \quad \leftarrow \beta_j^t - \eta \nabla RSS(\beta_j^t)$$

- We will converge when:
  - Fixed number of iteration.
  - Predefined threshold such as ϵ

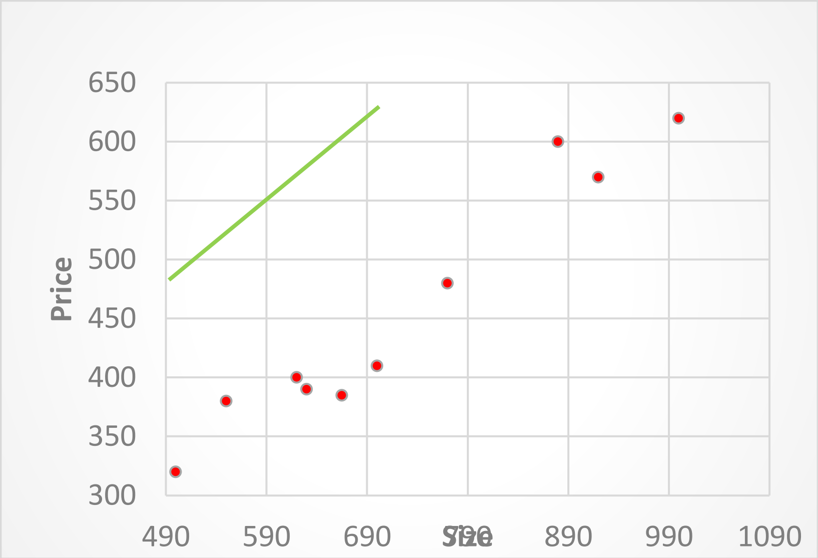


Gradient Descent Algorithm

$$\text{RSS}\ (\beta_0, \beta_1) = \sum_{i=1}^{N} (y_i - (\beta_0 + x_i\beta_1))^2$$

- Set gradient =0:

$$\nabla \text{RSS}\ (\beta_0, \beta_1) = \begin{bmatrix} \dfrac{\partial \text{RSS}\ (\beta_0, \beta_1)}{\partial \beta_0} = -2 \displaystyle\sum_{i=1}^{N} (y_i - (\beta_0 + x_i\beta_1) \\ \dfrac{\partial \text{RSS}\ (\beta_0, \beta_1)}{\partial \beta_1} = -2 \displaystyle\sum_{i=1}^{N} (y_i - (\beta_0 + x_i\beta_1) \quad (x_i) \end{bmatrix}$$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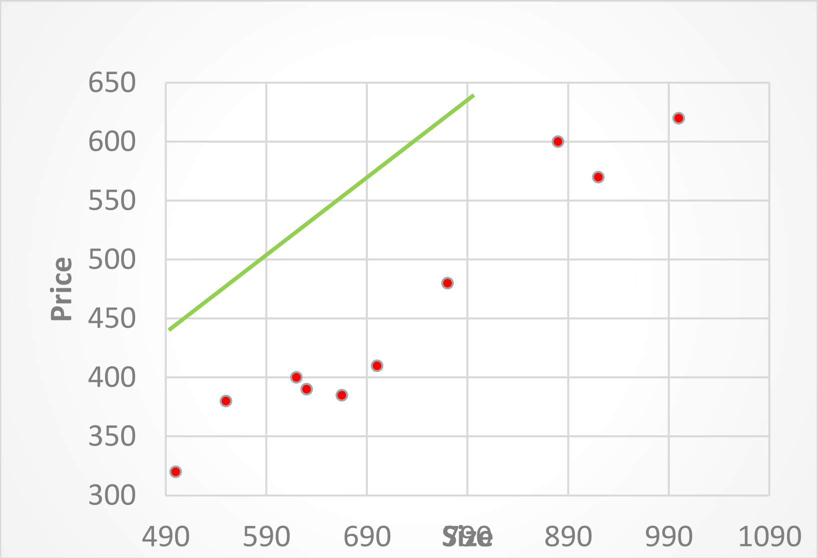
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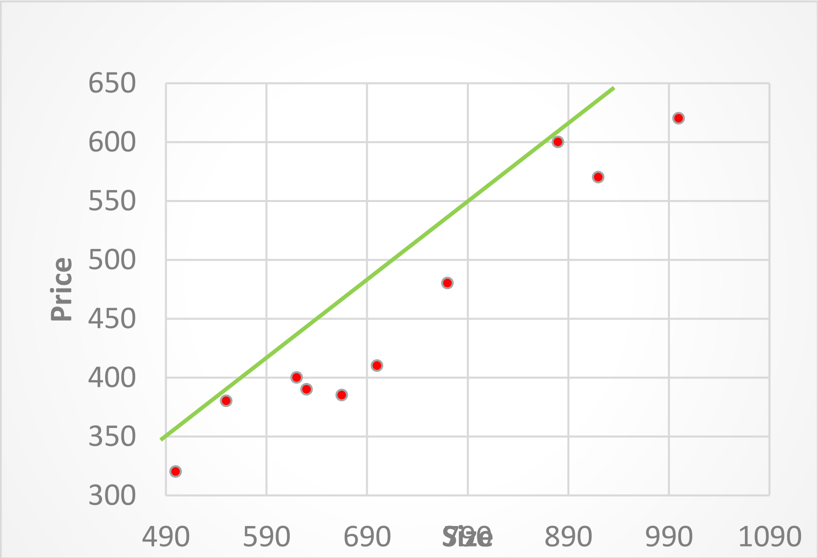
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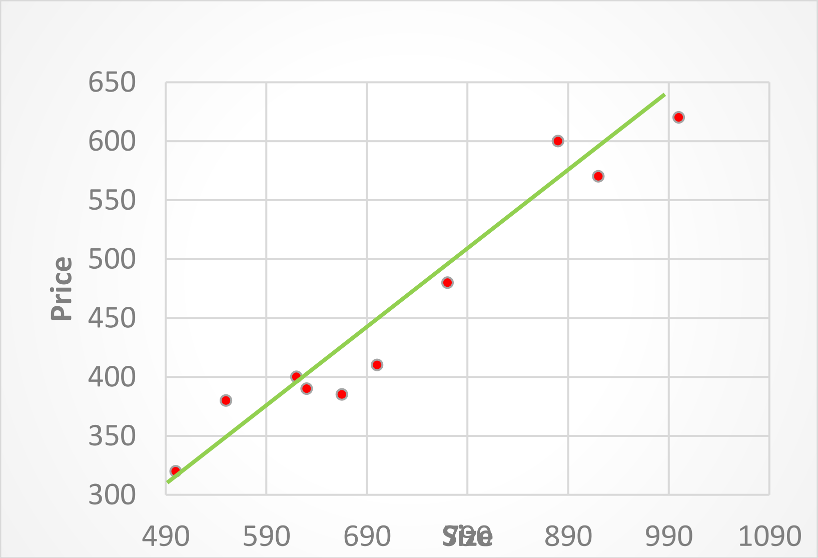
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1090
Size

Gradient Descent Algorithm

INPUT:

- A dataset $\mathcal{D}$ of training instances.
- A learning rate η or predefined threshold α
- A convergence threshold ϵ.
- Maximum number of iterations.
- β random starting point.

Repeat

For each β_j in β do

$$\beta_j^{t+1} \leftarrow \beta_j^t - \eta \nabla RSS\left(\beta_j^t\right) \quad \text{(where } 0 \le j \le 1\text{)}$$

Until convergence occurs

$$\boxed{\parallel \nabla RSS\left(\beta^t\right) \parallel < \epsilon}$$

INPUT:

- A dataset $\mathcal{D}$ of training instances.
- A learning rate η or predefined threshold α
- A convergence threshold ϵ.
- Maximum number of iterations.
- $g(\beta)$ β random starting point.

Repeat

For each β_j in β do

$$\beta_j^{t+1} \leftarrow \beta_j^t - \eta \nabla RSS\left(\beta_j^t\right) \quad \text{(where } 0 \le j \le 1\text{)}$$

Until convergence occurs

β

Multiple Linear Regression

office ID	Size	Floor	Energy Rating	Far from downtown	……..	Price $
1	500	4	C	1		320
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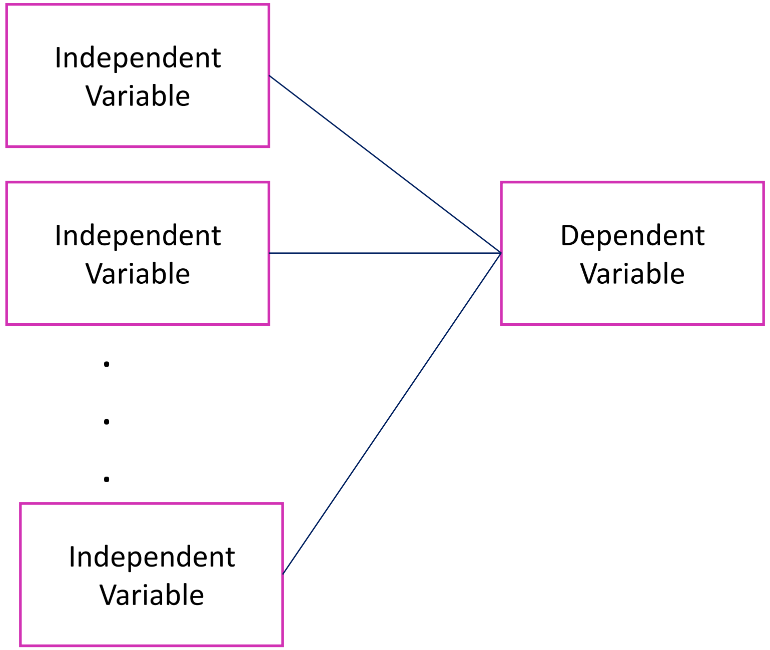
7	770	10	B	2		480
8	880	12	A	2		600
9	920	14	C	3		570
10	1000	9	B	1		620

Multiple Linear Regression

- Multiple regression is an extension to simple linear regression.



ID	x^1	x^2	...	x^D	Y
1					
2					
.					
.					
i					
.					
.					
.					
.					
N					

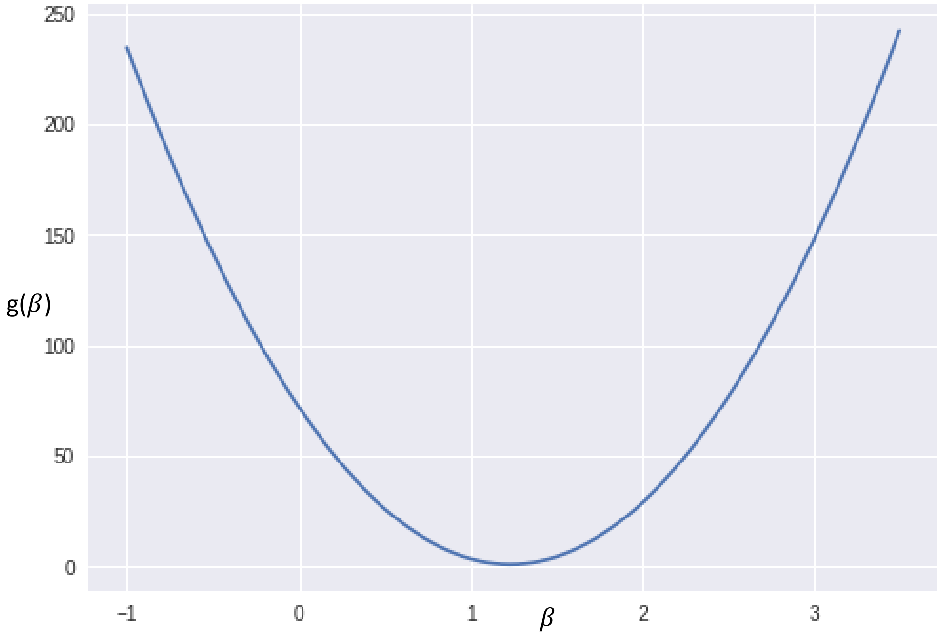
Multiple Linear Regression

Suppose that we have N observations $(x_1, y_1), \dots, (x_N, y_N)$, where $x_i = (x_1^{(1)}, x_2^{(2)}, \dots, x_i^{(j)}, \dots, x_N^{(D)})$ is a D-dimensional vector where $0 \le j \le D$ and $1 \le i \le N$, then

$$y_i = \beta_0 x_i^{(0)} + \beta_1 x_i^{(1)} + \beta_2 x_i^{(2)} + \cdots + \beta_j x_i^{(j)} + \cdots + \beta_D x_i^{(D)} + \varepsilon_i$$

- $x_i^{(j)}$: jth feature of ith data point.
- β_j: jth regression coefficient.

Gradient Descent Algorithm



INPUT:

- A dataset $\mathcal{D}$ of training instances.
- A learning rate η or predefined threshold α.
- A convergence threshold ϵ.
- Maximum number of iterations.
- β random starting point.

Repeat

For each β_j in β do

$$\beta_j^{t+1} \leftarrow \beta_j^t - \eta \nabla RSS\left(\beta_j^t\right) \text{ (where } 0 \leq j \leq D)$$

Until convergence occurs

$\boxed{\parallel \nabla RSS\left(\beta^t\right) \parallel < \epsilon}$

Multiple Linear Regression Interpretation

- Simple linear Regression

$$\hat{Y} = \widehat{\beta_0} + \widehat{\beta_1}X$$

- Multiple Regression Estimation

$$\hat{Y} = \widehat{\beta_0} + \widehat{\beta_1}X + \widehat{\beta_2}X$$

Rewrite in Matrix Notation

The regression equation can be expressed in a matrix format

Coefficient Estimation

- Remember, the cost for a given line

$$\text{RSS }(\beta_0, \beta_1) = \sum_{i=1}^{N} (y_i - \widehat{y_i})^2$$

- Let's generalize it,

$$\text{RSS }(\beta) = (Y - X\beta\,)^2$$

Coefficient Estimation

- Gradient of RSS

$$\nabla \mathrm{RSS}(\beta) = \nabla[(Y - X\beta)^T \ (Y - X\beta)]$$

When to use ..

Suppose we have N training examples, D features..

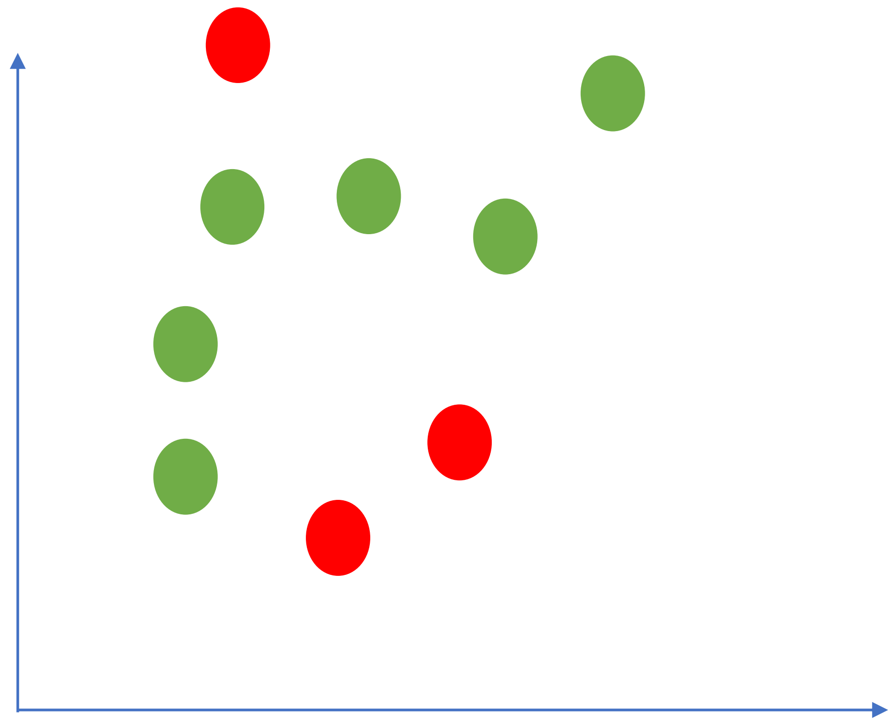
- Gradient Descent
  - Need to choose step size.
  - Needs many iterations.
  - Works well even with large D

- Normal Equation
  - No step size
  - No need to iterate
  - Slow if D is very large

Overfitting

- **Overfitting (high variance)**: where the model performs well on training data but does not generalize well to unseen data.

- **Underfitting (high bias)**: the model does not perform well on both training and testing data.



Ridge Regression

$$\text{Cost} = \sum_{i=1}^{N}(y_i \quad - \hat{y_i})^2 + \lambda \, \|\beta\|_2^2$$

Evaluating the Performance of Model

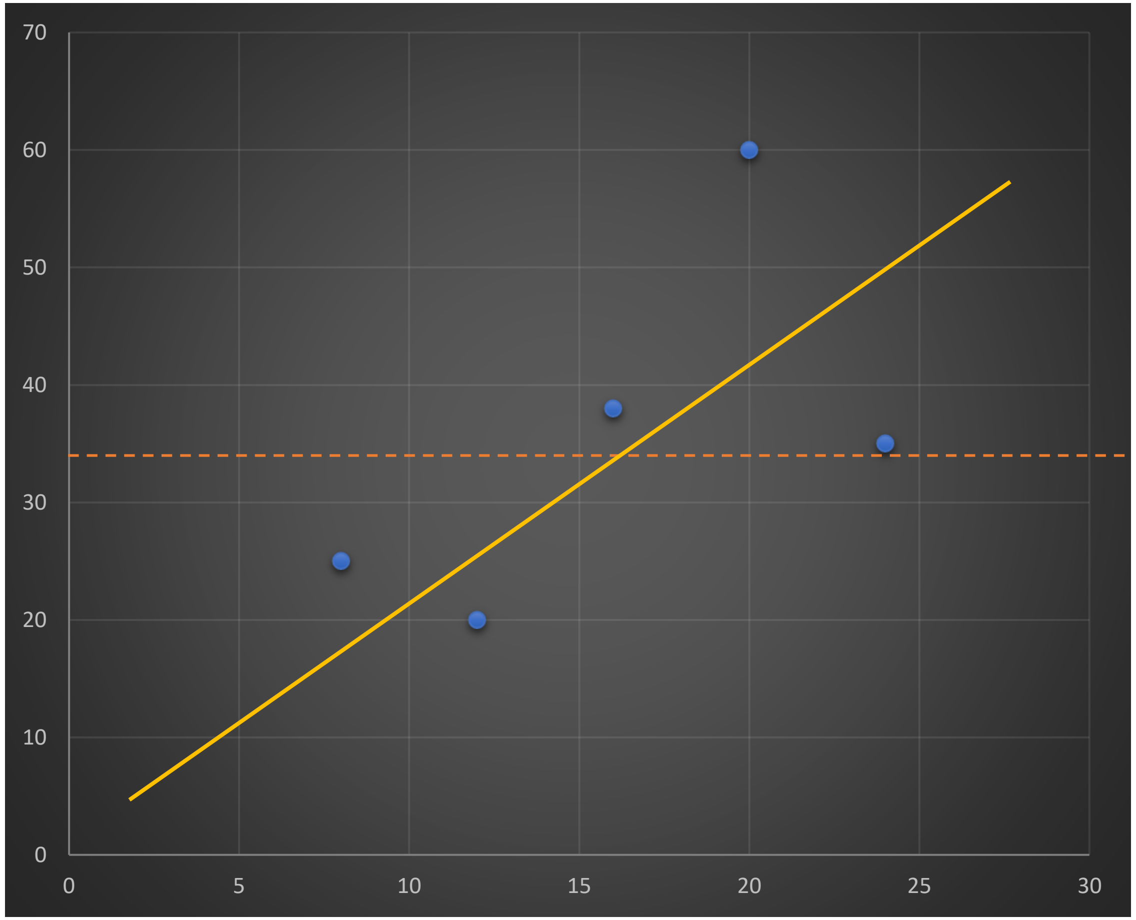
- Mean Square Error (MSE)

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

- Root mean square error (RMSE)
  - is the square root of the variance of the residuals.

$$RMSE = \sqrt{MSE}$$

R-squared



$$R^2 = \frac{\text{var(mean)} - \text{var(regression)}}{\text{var(mean)}}$$

R-squared

- What Is R-squared?
  - R-squared is a statistical indicates the percentage of the variance in the dependent variable that the independent variables explain collectively.

$$R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

  - R-squared is always between 0 and 1
    - 0 indicates that the model explains none of the variability of the response data around its mean.
    - 1 indicates that the model explains all the variability of the response data around its mean.

Adjusted R-squared

Adjusted R^2 tells you the percentage of variation explained by only the independent variables that actually affect the dependent variable.

$$\text{Adjusted } R^2 = 1 - \frac{SSE}{SST}\left(\frac{N-1}{N-p-1}\right)$$

- N: is the number of observations.
- p: is the number of independent variables.
- SSE: Error sum of squares
- SST: Total sum of square

Multiple Linear Regression

- In multiple regression, each coefficient is interpreted as the estimated change in the dependent variable corresponding to a one unit change in a independent variable when all other independent variables are held constant.

- Things to consider:
  - Multicollinearity.

Multicollinearity

- Multicollinearity: is whenever an independent variable is highly correlated with one or more of the other independent variables in a multiple regression equation.

- Consider the following regression

$$Salary_i = \beta_0 + \beta_1(Age_i) + \beta_2(Years\ of\ Experience_i) + \varepsilon_i$$

Multicollinearity

- How does it effect my model?
  - The coefficients become very sensitive to small changes in the model.
  - Multicollinearity reduces the precision of the estimate coefficients, which weakens the statistical power of your regression model.

Causes for Multicollinearity

- Human collection.

  - Including a variable in the regression that is a combination of two other variables.
    - For example, including “total investment income” when total investment income = income from stocks and bonds + income from savings interest.

  - Including two identical (or almost identical) variables.
    - For example, weight in pounds and weight in kilos.

Multicollinearity

- Detection:
  - Correlation
    - Check correlation between all pairs of independent variables (X).

Multicollinearity

- If the correlation coefficient, r, is exactly +1 or -1,
  - this is called perfect multicollinearity.
    - Near Singular Matrix

- If r is close to or exactly -1 or +1,
  - You could use one of the methods to deal with multicollinearity mentioned later in the slides.

How to Deal with Multicollinearity

- Do nothing
  - If correlation is not extreme.
  - If correlated variables are not of interest to study the equation

- Remove some of the highly correlated independent variables.
  - If the variables are providing the same information

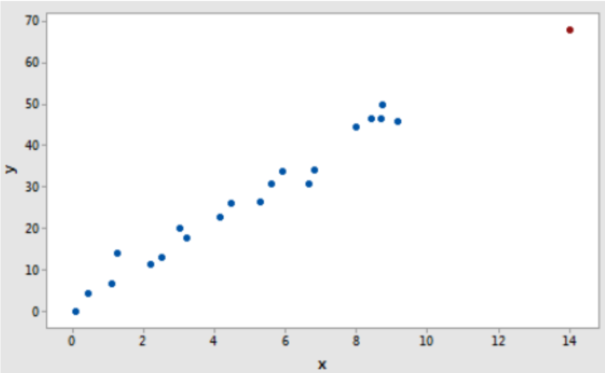
How to Deal with Multicollinearity

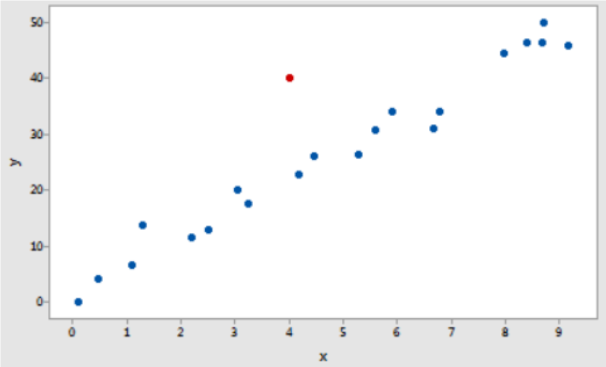
- Combine the independent variables
  - such as adding them together.
    - i.e. seniority score combines both Age and Experience

- Perform an analysis designed for highly correlated variables, such as principal components analysis.

Some insights

- Outlier data points
  - is a data point whose dependent variable (y) does not follow the general trend of the rest of the data.
- High leverage data points
  - A data point has high leverage if it has "extreme" independent variable (x) values.
- Influential data points
  - A data point is influential if it influences any part of a regression analysis,
    - such as the predicted responses, the estimated slope coefficients, or the hypothesis test results.


y
x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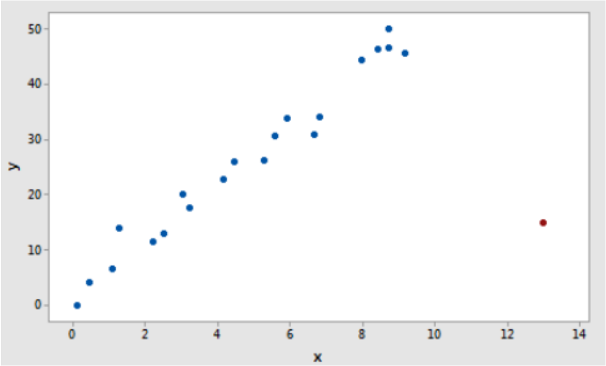
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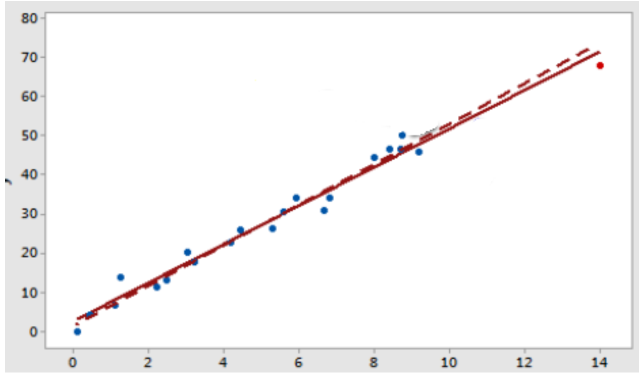

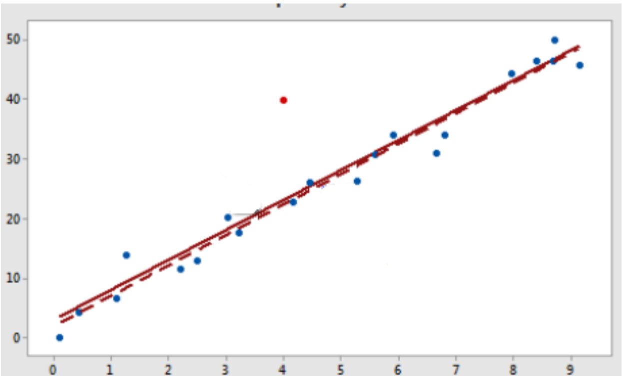

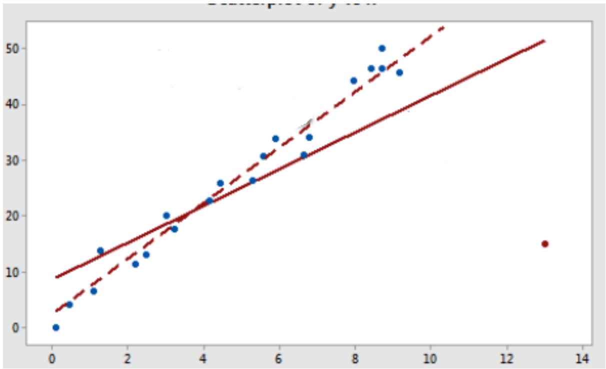

Recommended Reading

- **Regression Analysis by Example,** 5th Edition, Samprit Chatterjee and Ali S. Hadi (Note: hard copy is available at the library)
  - Read from page 25 to page 36.
  - Read from page 57 to page 69.
  - Read from page 93 to page 98.

- **The Elements of Statistical Learning** by Trevor Hastie, Robert Tibshirani, Jerome Friedman (Note: hard copy is available at the library)
  - Read from page 43 to page 55.

- **Python Machine Learning** by Sebastian Raschka & Vahid Mirjalili (2nd edition) (Skip the python codes if you are not interested)
  - Read from page 309 to page 347.

- **Machine Learning A Probabilistic Perspective** by Kevin P. Murphy (Note: hard copy is available at the library)
  - Read from page 219 to page 232.

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